CS348 – Project – Stages 2 and 3

Stage 2 Due Date: 3/28/2026

Stage 3 Due Date: 5/1/2026

Stage 2 deliverables:

1. Database design (e.g., database tables, primary keys, and foreign keys for a relational database). ERD is not required.
2. A 5- to 15-minutes demo for Requirements 1 and 2 (described in stage 1).
   a. Show how you can insert, update, and delete data on one of your tables.
   b. Show how you can filter data (e.g., specify a range of students ages and a range of GPAs) and display a report (e.g., a report showing the selected students). Display a report before and after some data is changed.
   c. The user interface of requirements 1 and 2 must retrieve data from the database if needed, such as when populating the items of a drop-down list. For example, a drop-down list or a list of check-boxes to select a particular course. Such a list must be built dynamically using data from the database. The courses must not be hard-coded in the interface. In your demo, show how you build those interface components dynamically (show code and/or query).

Stage 2 Grading Rubric

Stage 2 is 65% of the project grade

Deliverable	% of project grade
Database design	10%
Demo of requirement 1 is complete (point a above). Discuss the related source code of requirement 1.	25%
Demo of requirement 2 is complete (point b above). Discuss the related source code of requirement 2.	25%
User interface built with data from the DB (point c above).	5%
Total	65%

Stage 3 deliverables:

1. A final demo of your project. You will record a 5- to 10-minute demo. In the demo, **show parts of your code where you implemented the following course concepts**:
   a. How you protect your application from SQL Injection attacks (prepared statements, input sanitization, … etc.).

b. describe what indexes you have on your tables and what query(s) and report(s) those indexes support. For each index, show and discuss the query(s) that benefit from the index and where the queries are used (e.g., a specific report). Note that indexes will be covered in the next few weeks.
c. Transactions and your choice of isolation levels (transactions and concurrency will be covered in week 14 or 15). If your application is designed for a single user you may discuss a version where multiple users can access the same data concurrently.

2. A url to your application (if available) and your application's code (e.g., a GitHub link).

Extra credit:

1% of the course grade if you deploy your project (**database and application**) to Google Cloud Platform or any major cloud provider (e.g., Amazon AWS or Microsoft Azure). If you want to deploy your code to GCP you may use the app engine or compute engine (app engine is easier). If you do so, we expect a url to be available at the time of grading stage 3 (between 11:59 pm on 5/1/2026 and 11:59 pm on 5/15/2026). Remember to delete your instances and data after this date.

Stage 3 Grading Rubric

Deliverable	% of project grade
Discussion of SQL Injection protection method used in the application. Show related code.	7%
Discussion of Indexes. List queries that benefits from each index (be specific: report XXX or feat XXX). The indexes must be meaningful (important queries, reports, etc.). You must have a valid justification.	6%
Discussion of transactions, concurrent access to data, and the choice of isolation levels. If not applicable, 5% is added to indexes and 5% is added to demo.	6%
Discussion of how you used AI in your project (e.g., which tools you used and which tasks AI assisted with). Note that using AI is allowed in the CS348 semester project.	6%
Total	25%
Url to the application. A test of the live application (data is entered, updated, or deleted and a report reflects the changes made by the TA). The demo proves the database and app are hosted by some cloud provider.	Extra credit (1% of the course grade).

AI Acceptable Use Policy for the Semester Project

Students are allowed to use AI tools while working on the semester project. AI tools may help with productivity and learning when used responsibly. However, *students remain fully responsible for the work they submit and must understand and be able to explain all parts of their project.*

Allowed Uses

Students may use AI tools (such as coding assistants, chatbots, or AI-based development tools) for tasks such as:

- Generating or improving code snippets
- Writing SQL queries or debugging code
- Generating test cases or sample data
- Explaining programming concepts or error messages
- Assisting with UI design or project structure

Students must **review, verify, and modify AI-generated outputs as needed**.

Disclosure Requirement

Students must include a short section in their project code/documentation titled **"AI Usage"** describing:

- Which AI tools were used
- What tasks the AI assisted with
- How the student verified or modified the AI-generated output
- All of the above must be discussed in the recorded demo

Not Allowed

The following uses of AI are not permitted:

- **Submitting AI-generated work you do not understand.**
- Submitting a project largely generated by AI with minimal student contribution.
- Using AI to fabricate results and screenshots.
- Copying AI-generated code or designs without reviewing and verifying correctness.
- Using only information from AI without verifying it against the official documentation of the systems used in the project (e.g., web framework or database documentation).
- Using AI to circumvent course learning objectives.

Students must be able to **explain and justify all code, database design decisions, indexing strategies, security mechanisms (such as SQL injection protection), and transaction or isolation level choices used in their project.**

Responsibility

AI tools can assist development, but **the student is the author of the final submission** and is responsible for ensuring correctness, security, and academic integrity.

Failure to follow this policy may be treated as an **academic integrity violation**.

FAQ: Answers to Frequently Asked Questions

1. You may contact your TA to ask for help. However, the instructor and TAs will not be able to assist you in coding. You can ask for help by posting a question on Ed. Multiple students using the same tools are welcome to help one another by answering questions on Ed.
2. Can I use any database system? You are encouraged to use a relational database system (because we study those in more detail). However, you can use any database system that supports a query language, transactions, and indexes. Popular options for databases that you can use are: MySQL, Postgres, Oracle, SQL Server, MongoDB (document JSON database), and SQLite.
3. Some students' MySQL instances in Google Cloud got hacked before. Do your research and follow security best practices in the cloud provider you are using (if you select to do the extra credit).
4. Acknowledging minor bugs is encouraged. This is something you can discuss in the demo.
5. If prepared statements are not supported in your language/framework find another secure method for protecting your application against SQL injection attacks. If no methods are available, sanitize the user inputs and explain your approach in doing so.
6. Between stages 2 and 3, you are allowed to change your database/UI design and/or the language/database/frameworks you are using. Discuss those changes in your final demo in stage 3.
7. There is no minimum number of indexes. However, make sure to support the important queries with indexes if needed, such as your report, your login (if you have one), retrieving data for your interface (e.g., populating data items for a drop-down menu or a list of check boxes). Those are frequent queries and must be supported with indexes.